FORCe single-trial onset check

Fresh pilot using the revised implementation, continuous 1 Hz high-pass, fixed two-level decomposition, 500 Hz EEG and 20 Hz low-pass **after** FORCe. Compare two separate 1 s calls against one 2 s call; other settings are fixed.

Both participants, all 29 eligible recordings: **2,616 matched trials**. Each window method retained 2,617 trials. Gray example traces show the same input processed without FORCe.

Measurement (microvolts)	1 s	2 s
Before low-pass: median	7.974	3.503
Before low-pass: 95th percentile	20.623	9.435
Before low-pass: maximum	40.236	20.405
After low-pass: median	0.539	0.423
After low-pass: 95th percentile	1.397	1.049
After low-pass: maximum	3.330	2.469
Nearby before low-pass: median	3.595	3.363
Nearby after low-pass: median	0.509	0.429

Definition: each trial contributes its largest absolute channel change between -2 ms and 0 ms. Nearby steps summarize adjacent-sample changes within +/-50 ms, excluding that pair. These are signal-change measurements, not pure artifact amplitudes.

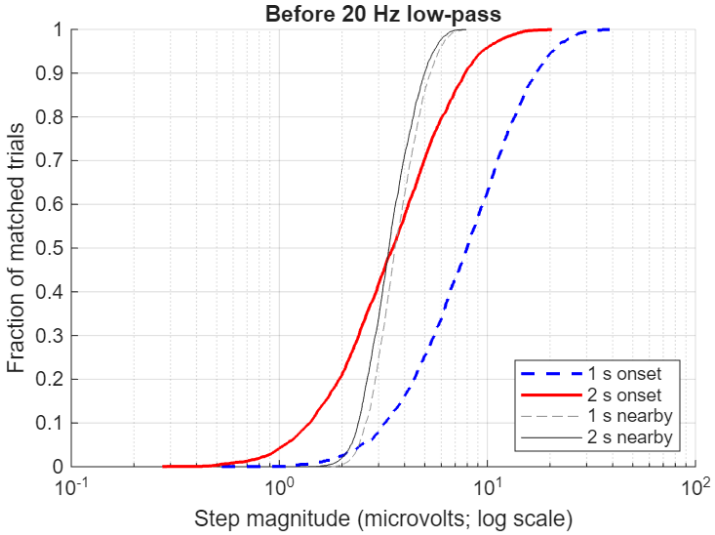
Finding: a clear excess exists at the join before low-pass filtering. The final-signal steps are much smaller. Two-second cleaning also changes broader waveforms in some examples, so a smaller join does not establish better artifact removal or neural-signal preservation.

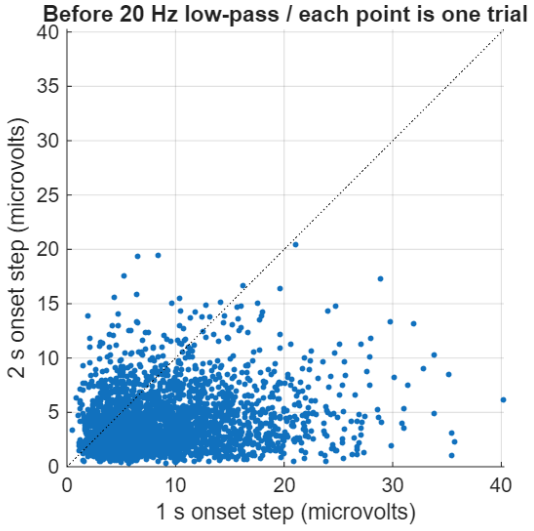
Examples: median, 75th percentile, 95th percentile, then the three largest pre-filter one-second steps. Channels are selected by that same rule; they are not necessarily Cz or FCz. Blue dashed = 1 s; red solid = 2 s; gray = no FORCe.

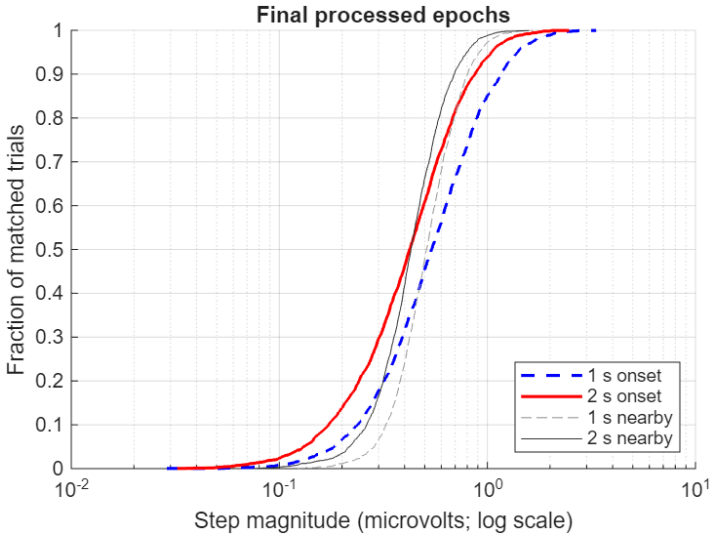
No-FORCe QC 1 passes the final 100 microvolt limit; QC 0 fails. Two extreme examples have QC 0 and their gray traces are contextual references. All window comparisons use trials retained by both FORCe methods. This check does not measure classifier performance.

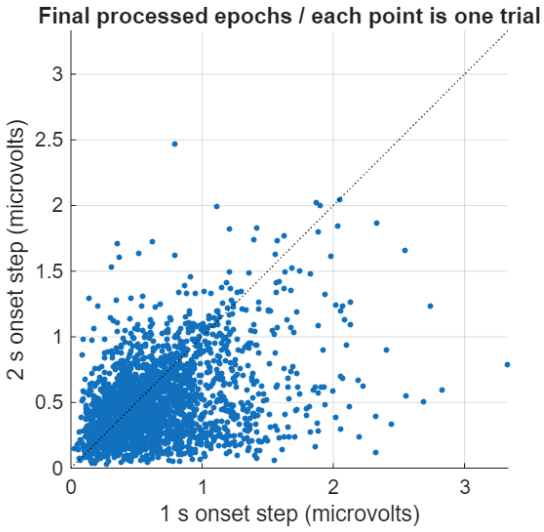
All matched trials

Revised library / 2616 matched trials / largest absolute channel step from -2 ms to 0 ms

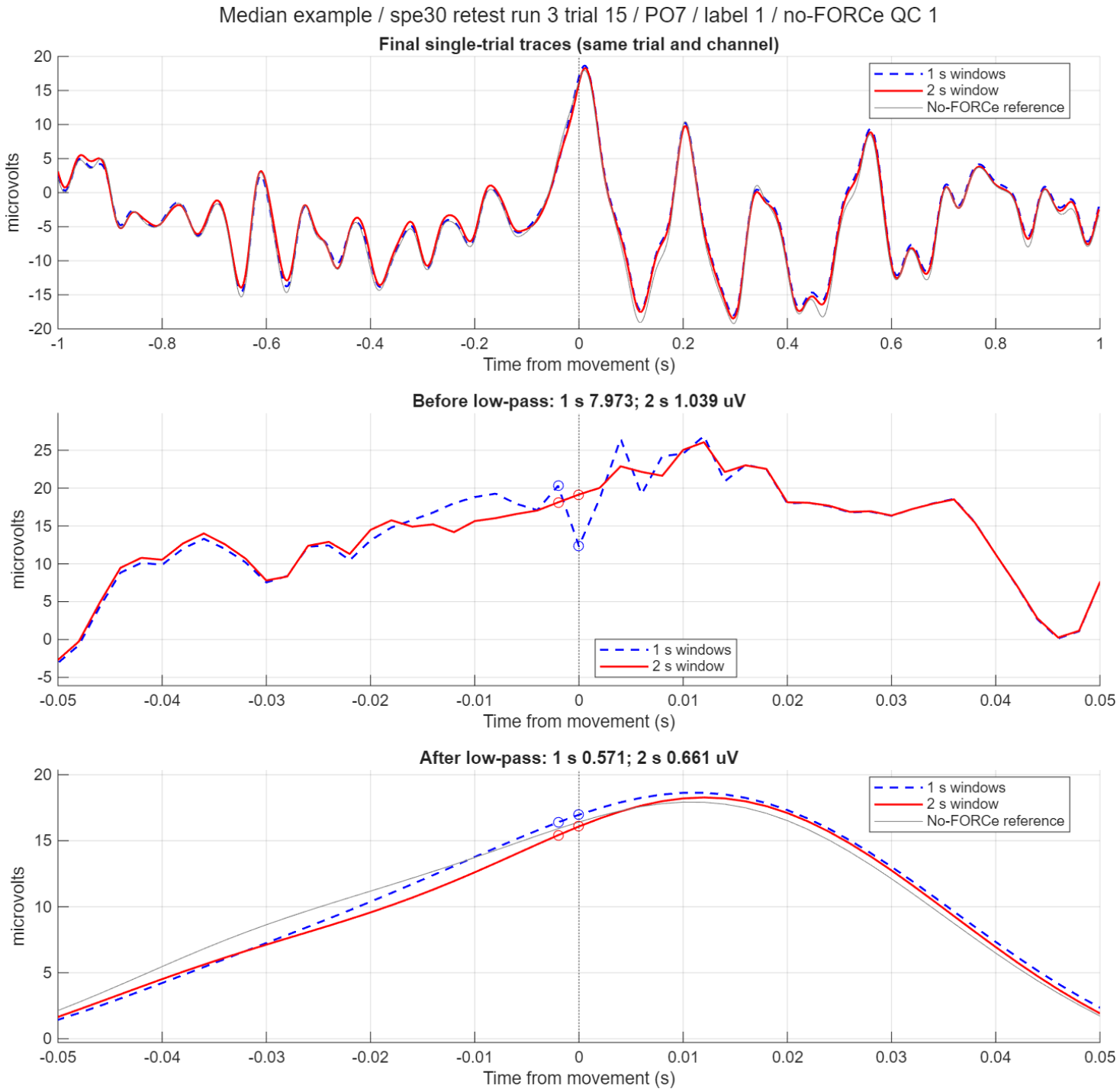




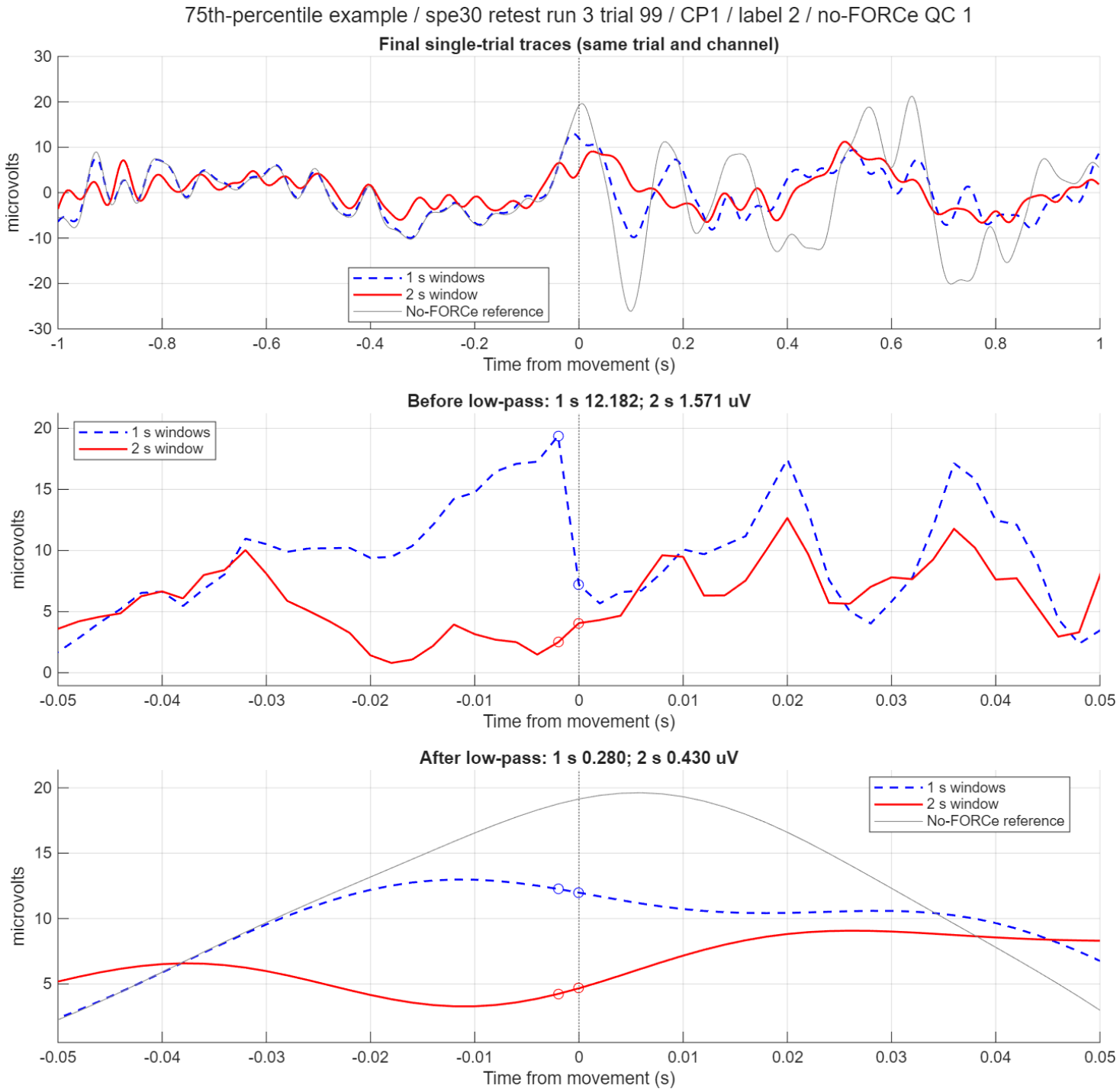




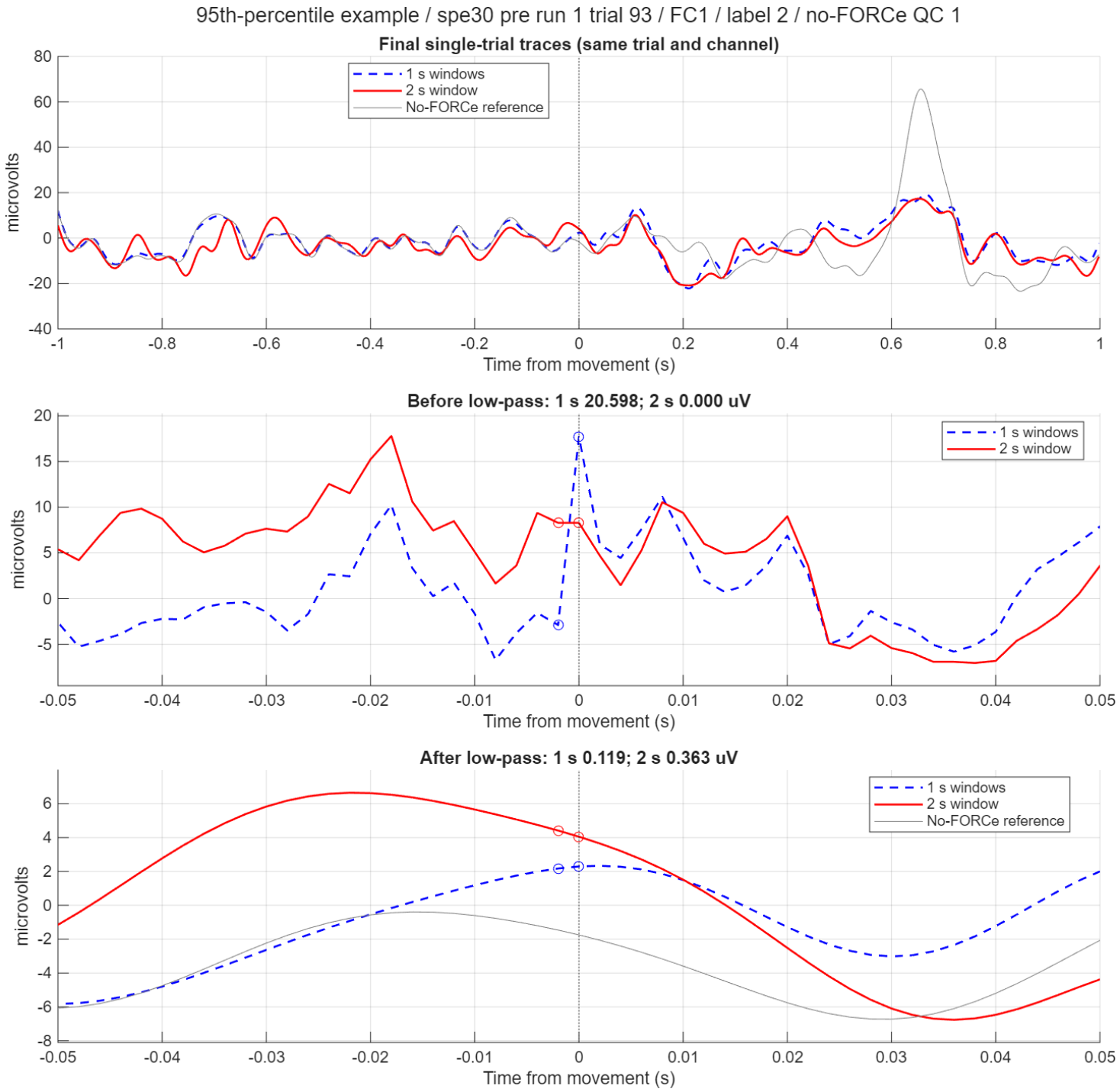
Median example



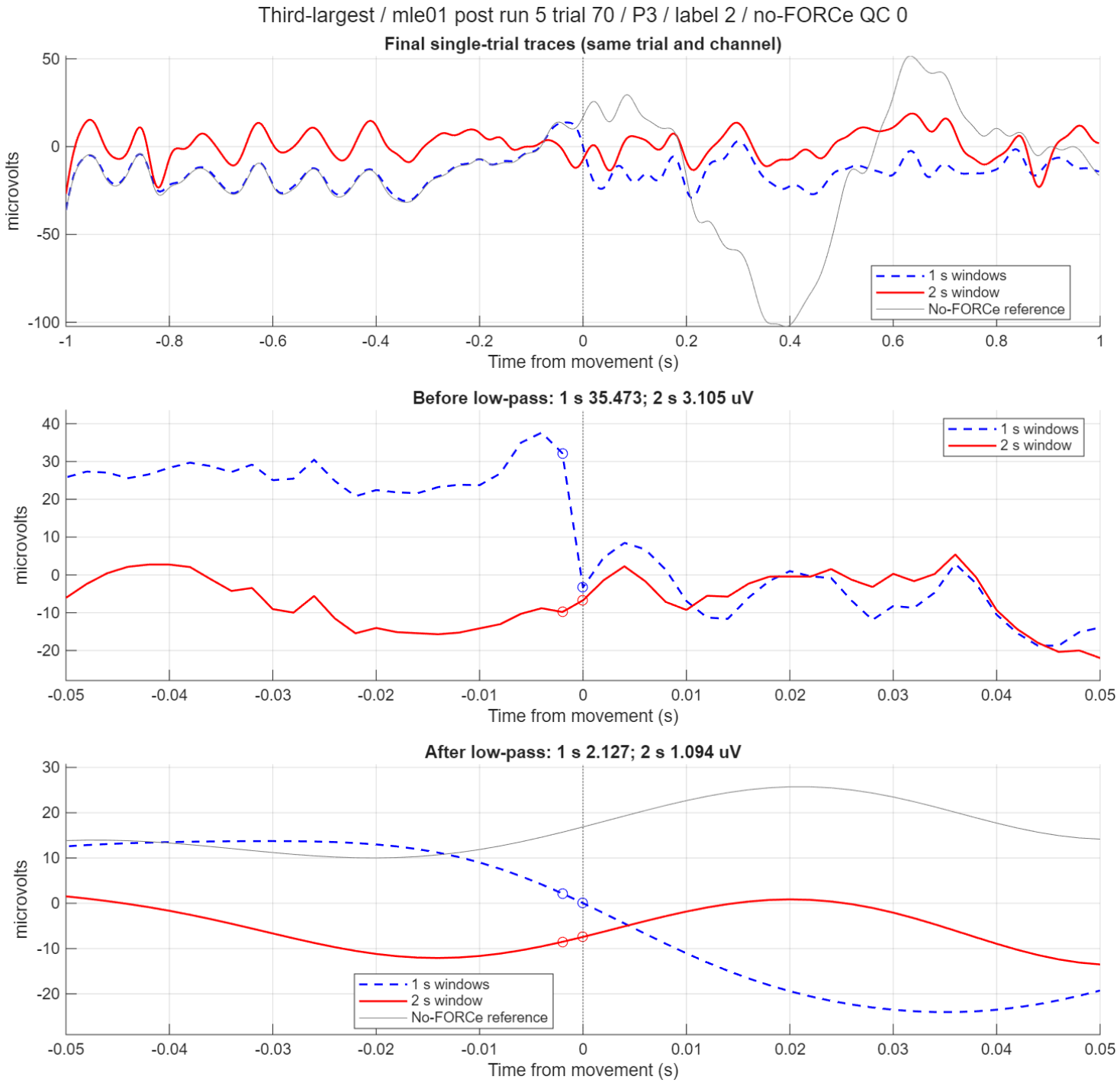
75th-percentile example



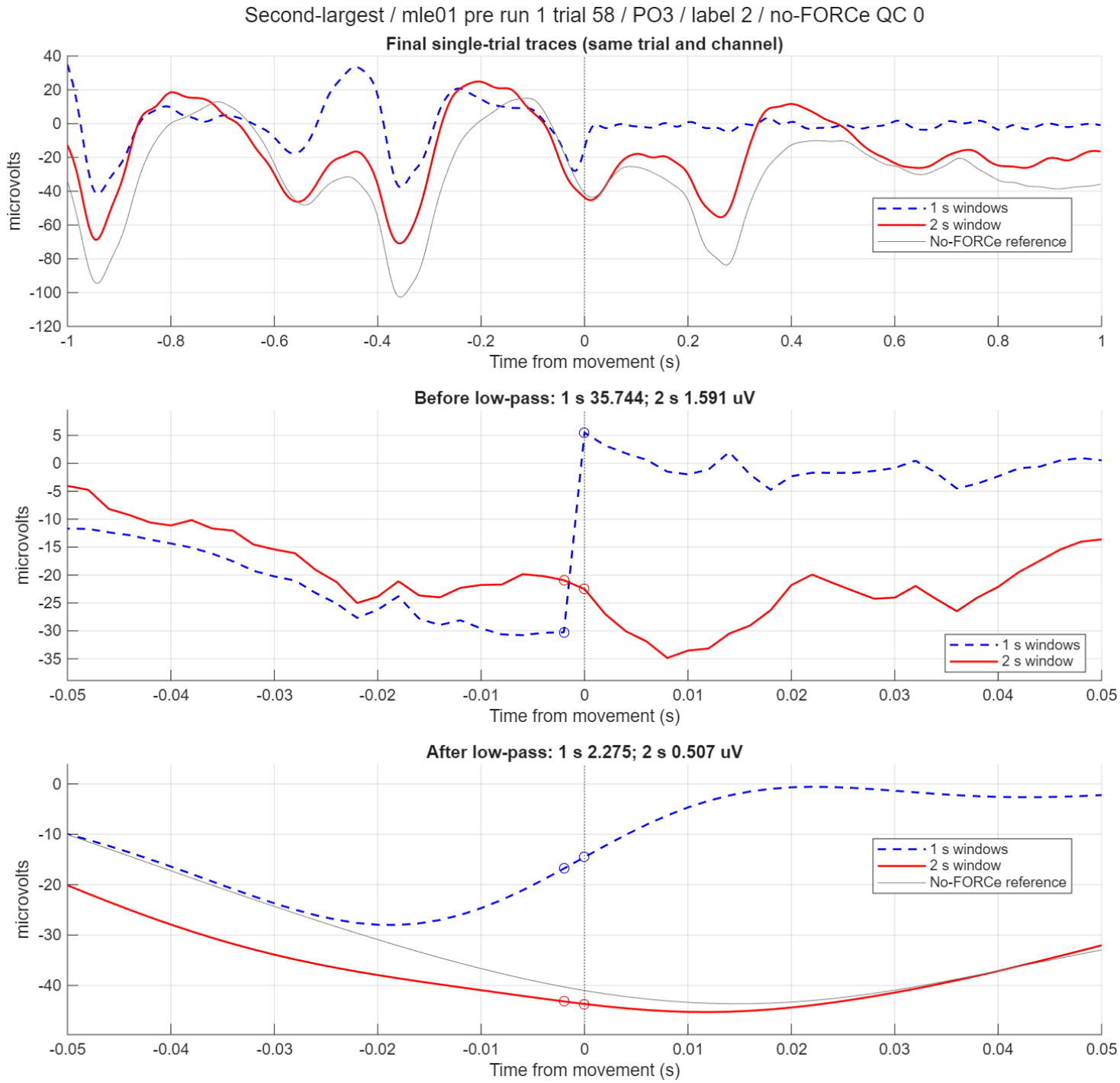
95th-percentile example



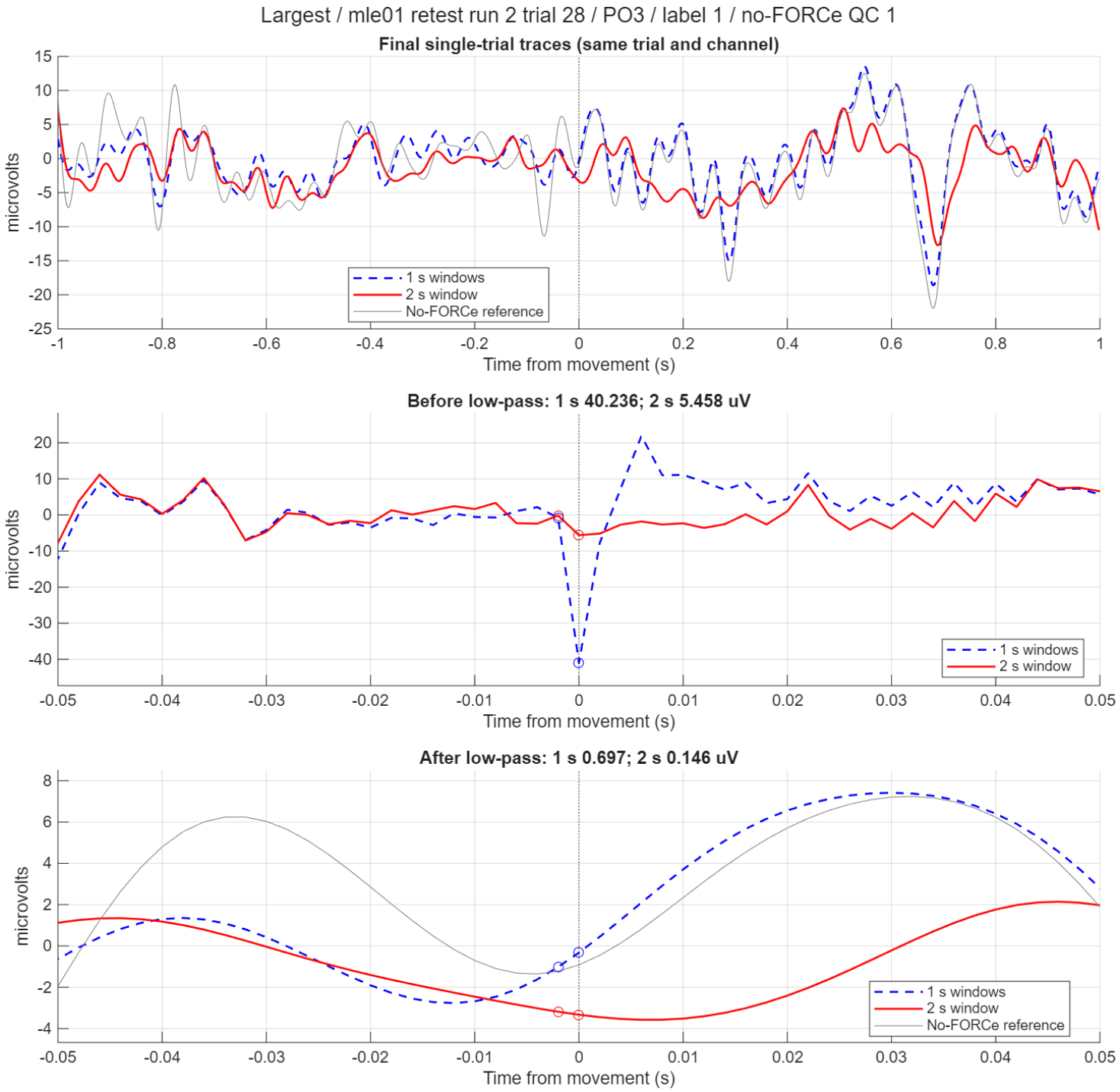
Third-largest



Second-largest



Largest


Largest / mle01 retest run 2 trial 28 / PO3 / label 1 / no-FORCe QC 1
Final single-trial traces (same trial and channel)
microvolts
Time from movement (s)
1 s windows
2 s window
No-FORCe reference
Before low-pass: 1 s 40.236; 2 s 5.458 uV
microvolts
Time from movement (s)
1 s windows
2 s window
After low-pass: 1 s 0.697; 2 s 0.146 uV
microvolts
Time from movement (s)
1 s windows
2 s window
No-FORCe reference